



Eduqat Builds Highly Scalable AI-Powered Online Learning Platform on AWS

Discover how the Indonesia-based EdTech company is transforming the country’s education sector using AWS. Its AI-powered online learning platform is supporting the government’s mission to boost workforce employability.

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100% availability

Uninterrupted access for learners and



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Up to 5,000

Active concurrent users supported in a test environment



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15 Minutes



Overview

In the wake of rising unemployment and the pressing need for skills enhancement among Indonesia's workforce, the government launched the ambitious Prakerja program in 2019. Amid this backdrop, [Eduqat](#) emerged—a beacon of innovation in the country's education technology (EdTech) landscape.

As a key player under this national initiative, Eduqat utilizes artificial intelligence (AI) to empower educators to easily build comprehensive online courses. This marks a significant leap toward redefining the future of education in Indonesia.

As of January 2024, Eduqat had attracted 200 schools and educators, as well as amassed a waiting list of 2,000 learners during the beta phase, thus demonstrating its rapid growth and innovation. The company's approach to education is powered by Amazon Web Services (AWS), thus providing the scalability and reliability that drive the success of Eduqat's online Learning Management Systems (LMS).



Opportunity | Building a Scalable, Secure Infrastructure for Indonesia's Digital Education Drive

Eduqat was established in 2023 to address the growing demand for online, easy-to-use learning platforms in Indonesia. Before 2023, the company operated under a different name as an online learning marketplace that connected learners with offline vocational classes, workshops, and webinars. The company also used an open-source web content management system (CMS) for its infrastructure. This system, while adequate for simple directory-related tasks, soon experienced issues related to scalability, resilience, and data security.



For instance, during the pandemic, the company experienced a 5,000 percent surge in demand for its existing online learning platform, which its servers were unable to adequately handle. Moreover, there were data security-related concerns, since the company handled sensitive personal information from learners and educators alike.

These concerns became especially critical after joining the Prakerja program in 2020, which required a system capable of reliably and securely supporting up to 100,000 learners per month. Since the company's existing online learning platform did not meet said requirements, a new solution had to be devised.

Eduqat, in its current form, is a secure, customizable, and user-friendly online learning platform that allows educators to create, manage, and sell their online courses. Eduqat has also integrated AI technologies to help educators build landing pages, plan curriculums, and create quizzes. The goal of Eduqat is to empower educators and provide learners with a seamless online learning experience.

Solution | Transforming Digital Learning through AI and AWS

In 2023, Eduqat worked with AWS to migrate and build from an external LMS to a cloud infrastructure, significantly enhancing scalability, reliability, and data security. Using AWS, the company could now manage the intensive computing demands of AI/Machine Learning (ML) projects. As a result, educators on Eduqat could quickly create custom white-label websites and generate course content within 15 minutes. Moreover, they could grade assessments and provide tailored feedback promptly within two minutes.

The company implemented [Amazon Elastic Compute Cloud \(Amazon EC2\) Auto Scaling](#), an automated cloud compute scaling service, for dynamic scaling. This delivered consistent uptime and seamless handling when traffic volumes increased. Eduqat also used [Amazon Elastic Container Service](#) (Amazon ECS), a fully managed container orchestration service, to automatically deploy, manage, and scale its online services based on user volumes without unnecessary resource allocation. With both services, the company aims to serve up to 300,000 users with 100 percent uptime, and create reliable and positive learning experiences throughout the beta phase.

Utilizing the combination of [Amazon DynamoDB](#), a serverless, fully managed database, and [Amazon Relational Database Service for PostgreSQL](#) (Amazon RDS for PostgreSQL), a fully managed database service, Eduqat could scale its database performance by several hundredfolds. The company also relied on [AWS Lambda](#), a serverless, event-driven compute service, to execute code and automate response requests as needed. With fully managed services in place, Eduqat no longer had to provision for or manage infrastructure, freeing the IT team to focus on innovation and growth.

For security, Eduqat activated [AWS Identity and Access Management](#) (IAM), an identity and access management service, to oversee user and service access across all AWS services. This ensured only authorized access to critical resources, safeguarding against unauthorized entry and data breaches.

The company also deployed [Amazon Simple Storage Service](#) (Amazon S3), an object storage service that offers industry-leading scalability, data availability, security, and performance, to operate and scale more than 500 terabytes worth of educational content in the cloud. With this service, Eduqat could now adjust its database resources as needed. During the beta phase, users had uninterrupted access to the full suite of learning materials 24/7.

Finally, with [Amazon CloudFront](#), a low-latency content delivery network service, learners can view or download educational content significantly faster compared to the previous open-source CMS.

Outcome | Driving an AI-Powered Learning Vision across Southeast Asia

Aside from conducting acquisitions, Eduqat has plans to collaborate with 50 EduTech service providers, each comprising 20 teachers and 10 programs. Eduqat is also targeting up to 60,000 participants as part of the Prakerja program, then expanding its reach to the rest of Southeast Asia within the next two years. Finally, by the end of 2024, Eduqat plans to launch an AI-powered virtual tutor to enhance personalized learning anytime, anywhere, and improve educational outcomes.

Overall, this strategic move toward leveraging AI, supported by AWS, underscores Eduqat's commitment to fostering innovation and accessibility in learning across Indonesia and the region.

“By working with AWS, Eduqat is poised to lead in digital learning across Indonesia and beyond, especially through the Kartu Prakerja program. The flexibility and scalability provided by AWS lets us fast-track the development and integration of AI technologies, empowering our educators to deliver more personalized learning experiences. This effort highlights our commitment to driving innovation and making adaptive education widely accessible,” said Jourdan Kamal, Chief Executive Officer and Co-founder of Eduqat.

Learn More

To learn more, visit aws.amazon.com/education.

About Eduqat

Eduqat is an Indonesia-based education technology company and one of the top five Learning Systems under Indonesia's Prakerja program. Founded in 2023, Eduqat allows educators to build entire websites



and course curriculums using artificial intelligence, and then sell their courses, events, and coaching sessions. While Eduqat was still in beta as of January 2024, the company had already onboarded up to 200 educators with over 2,000 learners on an active waiting list across Indonesia. As part of the Prakerja program, Eduqat is proud to support the empowerment of Indonesians by enhancing their skills and competitiveness in the job market. This program has so far benefited 17.5 million individuals across 514 districts and cities throughout Indonesia. Over three years, the Prakerja Program has provided inclusive training opportunities, with 51 percent of participants being women and two percent coming from underserved districts and cities. Overall, it has led to a 21 percent increase in monthly income and improved job prospects for its recipients.

AWS Services Used

Amazon EC2 Auto Scaling

Amazon EC2 Auto Scaling helps you maintain application availability and lets you automatically add or remove EC2 instances using scaling policies that you define.

[Learn more »](#)

Amazon Elastic Container Service

Amazon Elastic Container Service (ECS) is a fully managed container orchestration service that helps you to more efficiently deploy, manage, and scale containerized applications.

[Learn more »](#)

Amazon DynamoDB

Amazon DynamoDB is a serverless, NoSQL database service that enables you to develop modern applications at any scale.

[Learn more »](#)

AWS Lambda



AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

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2024

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